

AYE

AYE TECH HUB

Engineering. Knowledge. Innovation.

INDUSTRIAL MAINTENANCE MANUAL

A Complete Engineering Reference Guide

TOPICS COVERED

- Preventive Maintenance
- Predictive Maintenance
- Reliability-Centered Maintenance
- Work Order Management & CMMS
- Lubrication & Tribology
- Mechanical Systems
- Hydraulic & Pneumatic Systems
- Electrical Maintenance
- Maintenance Safety (LOTO/PTW)
- Spare Parts Management
- KPIs & Maintenance Metrics
- Troubleshooting Guide

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01 Introduction to Industrial Maintenance

Industrial maintenance is the systematic set of activities aimed at keeping plant equipment, machinery, and infrastructure in safe, reliable, and productive operating condition. An effective maintenance strategy directly impacts equipment uptime, product quality, operational safety, and overall plant profitability.

Maintenance Philosophy Overview

Modern industrial maintenance has evolved from purely reactive "fix-it-when-it-breaks" approaches to data-driven, reliability-focused strategies. The four primary philosophies are applied based on equipment criticality, failure consequences, and cost-benefit analysis.

Maintenance Type	Trigger	Best Applied When	Relative Cost
Corrective (BM)	Equipment failure	Non-critical, cheap assets	Low upfront, High total
Preventive (PM)	Calendar / usage interval	Known wear-out failure patterns	Medium
Predictive (PdM)	Condition monitoring data	Critical rotating equipment	High upfront, Low total
Proactive / RCM	Failure mode analysis	Safety-critical plant systems	Highest — best ROI

The True Cost of Poor Maintenance

Studies by the US Department of Energy estimate that reactive-only maintenance costs 3–5× more per machine than a planned preventive programme due to unplanned downtime, secondary damage, emergency labour, and expedited spares.

■ Engineering Note

World-class facilities target Overall Equipment Effectiveness (OEE) above 85%.

Best-in-class maintenance ratio: Planned Work > 80% of total man-hours.

Maintenance costs should be 2–5% of Replacement Asset Value (RAV) annually.

Every \$1 invested in planned maintenance returns \$4–8 in avoided failure costs.

02 Preventive Maintenance Planning

Preventive Maintenance (PM) consists of scheduled, time-based or usage-based maintenance tasks performed to reduce the probability of equipment failure. PM tasks are defined by equipment manufacturer recommendations, historical data, and engineering judgement — then executed before failure occurs.

PM Task Categories

- **Inspection:** Visual checks, gauging, listening for abnormal sounds — non-intrusive, high frequency.
- **Lubrication:** Re-lubrication, oil changes, filter replacements on fixed intervals.
- **Cleaning:** Removal of contamination (dust, debris, process material) from mechanical and electrical components.
- **Adjustment:** Belt tension, alignment checks, valve packing, torque verification.
- **Replacement:** Planned swap of wear items (belts, filters, seals, consumables) at life-cycle intervals.
- **Testing:** Functional verification — safety devices, trip circuits, protection relays, interlocks.

PM Frequency Determination

Equipment Class	Recommended PM Interval	Key Tasks
Critical rotating (pumps, compressors)	Monthly	Vibration check, lubrication, alignment
Conveyors & material handling	Weekly	Belt tension, bearing temp, lubrication
Electrical panels & MCC	Quarterly	Thermography, torque check, insulation test
HVAC & cooling systems	Monthly	Filter change, coil clean, belt inspect
Safety valves & relief devices	Annual	Lift test, set-pressure verification
Fire & gas detection	Monthly	Functional test, sensor calibration

■ PM Programme Success Factors

Compliance rate target: > 95% of scheduled PMs completed on time.

Use failure history to optimise PM frequency — avoid over- and under-maintenance.

Standardise task procedures (SWPs) to ensure consistency across technician shifts.

03 Predictive Maintenance Technologies

Predictive Maintenance (PdM) uses condition monitoring technologies to detect deterioration in equipment before failure occurs. By trending key parameters (vibration, temperature, oil quality, electrical signature), engineers can schedule maintenance precisely when needed — maximising run time and minimising unnecessary work.

PdM Technology	Detects	Equipment	Frequency
Vibration Analysis	Imbalance, misalignment, bearing wear, looseness	Rotating machinery	Monthly
Infrared Thermography	Hot spots, overloaded circuits, heat patterns	Electrical, mechanical	Quarterly
Oil Analysis / Tribology	Wear metals, contamination, degradation	Gearboxes, engines	Quarterly
Ultrasonic Testing	Leaks (air, steam), bearing defects, arcing	Bearings, valves, electrical	Monthly
Motor Current Analysis	Rotor bar faults, eccentricity, load variation	Electric motors	Monthly
Laser Alignment	Shaft misalignment (angular, parallel)	Couplings, shafts	As-needed

Vibration Severity — ISO 10816 Guidelines

ISO 10816 classifies vibration severity into four zones for rotating machinery. RMS velocity is the most common parameter used for continuous monitoring:

- **Zone A (Good):** < 2.3 mm/s RMS — newly commissioned machinery, no action required.
- **Zone B (Acceptable):** 2.3–4.5 mm/s RMS — suitable for long-term continuous operation.
- **Zone C (Alert):** 4.5–7.1 mm/s RMS — operate short-term only; plan maintenance.
- **Zone D (Danger):** > 7.1 mm/s RMS — damage risk; shut down and inspect immediately.

■ PdM Implementation Note

Begin with critical assets only — expand programme after establishing baselines.

Trend data over time is more valuable than a single reading.

Combine multiple PdM techniques for highest confidence in fault diagnosis.

04 Reliability-Centered Maintenance (RCM)

Reliability-Centered Maintenance (RCM) is a structured analytical process used to determine the maintenance requirements of any physical asset in its operating context. Originally developed by the US aviation industry (Nowlan & Heap, 1978), RCM is now the foundation of maintenance strategy in oil & gas, power generation, mining, and process plants.

The Seven RCM Questions

- **Function:** What is the asset required to do in its current operating context?
- **Functional Failure:** In what ways can it fail to fulfil each function?
- **Failure Mode:** What causes each functional failure?
- **Failure Effect:** What happens when each failure mode occurs?
- **Failure Consequence:** In what way does each failure matter (safety/environment/operational/economic)?
- **Proactive Task:** What can be done to predict or prevent each failure?
- **Default Action:** What should be done if no proactive task can be found?

FMEA — Failure Mode and Effects Analysis

Failure Mode	Effect	RPN*	Maintenance Action
Bearing wear (fatigue)	Vibration → unplanned stop	High	Vibration monitoring monthly
Seal leakage	Product loss, contamination	Med	PM seal inspection quarterly
Impeller erosion	Reduced flow, cavitation	Med	Ultrasonic thickness 6-monthly
Motor winding degradation	Overheating → burnout	High	Insulation test (megger) annually
Coupling misalignment	Vibration, bearing load	Med	Laser alignment at each outage

*RPN = Risk Priority Number (Severity × Occurrence × Detectability, scale 1–1000)

■ RCM Outcome

RCM analysis typically results in: 30–50% reduction in scheduled maintenance tasks.
 Remaining tasks are justified — each task prevents a real failure with real consequences.
 Software tools: SAP PM, Maximo, Meridium, Isograph Availability Workbench.

05 Work Order Management & CMMS

A Computerised Maintenance Management System (CMMS) is the digital backbone of any modern maintenance organisation. It manages work orders, tracks asset history, schedules PMs, controls spare parts inventory, and generates KPI reports — replacing paper-based systems and spreadsheets with a single source of maintenance truth.

Work Order Lifecycle

- **Identification:** Fault detected by operator, inspection, or PdM sensor alert.
- **Notification:** Work request raised in CMMS with description, location, and priority.
- **Planning:** Maintenance planner assigns craft, estimates labour hours, identifies spares.
- **Scheduling:** Work scheduled into maintenance window; resources confirmed available.
- **Execution:** Technician carries out work; completes job card with findings and time.
- **Completion:** Supervisor reviews, approves WO closure; failure cause code captured.
- **Analysis:** Data feeds into KPI reports, failure trends, and PM optimisation.

Work Order Priority Classification

Priority	Description	Response Time	Example
P1 — Emergency	Safety / production stop	Immediate	Fire system failure, major leak
P2 — Urgent	Imminent failure risk	4 hours	Bearing temperature alarm
P3 — High	Operational impact	24 hours	Pump performance degradation
P4 — Routine	Scheduled PM or minor work	1 week	Oil change, filter replacement
P5 — Improvement	No operational impact	Next outage	Modification, minor upgrade

■ CMMS Selection Criteria

Key features: WO management, PM scheduling, asset registry, inventory, reporting.

Leading platforms: SAP PM, IBM Maximo, Infor EAM, Fiix, UpKeep, Limble CMMS.

Target: planned work ratio > 80%; backlog < 2 weeks of crew capacity.

06 Lubrication & Tribology

Lubrication is responsible for reducing friction, wear, and heat in moving components. Studies show that 36–50% of premature bearing failures are caused by improper lubrication — either wrong lubricant type, incorrect quantity, or wrong interval. A well-managed lubrication programme is one of the highest-ROI maintenance activities.

Lubricant Selection Criteria

Component	Lubricant Type	ISO Viscosity Grade	Change Interval
Rolling element bearings	Grease (NLGI 2–3)	N/A	500–2000 hrs or manufacturer spec
Plain/journal bearings	Circulating oil	ISO VG 46–68	Annual with oil analysis
Gearboxes (enclosed)	Gear oil	ISO VG 220–680	6-monthly or oil analysis
Hydraulic systems	Hydraulic oil	ISO VG 32–46	Annual or 2000 hrs
Air compressors	Compressor oil	ISO VG 46–100	Per OEM spec (500–2000 hrs)
Chains & open gears	Spray lube / grease	NLGI 0–1	Weekly or condition-based

Grease Re-Lubrication Formula

The Mobil re-lubrication formula provides a bearing-specific re-greasing interval based on bearing bore diameter (d) and rotation speed (N):

Re-Lubrication Interval Formula

$$\text{Hours} = K \times [(14,000,000 / (N \times d^{0.5})) - 4d]$$

Where: K = 1.0 (ball bearings), 0.5 (roller bearings)

N = rotational speed (RPM), d = bore diameter (mm)

Example: 6208 bearing (40mm bore) at 1450 RPM = K=1.0 → ≈ 1,920 hrs

Reduce interval by 50% for: high temp (>70°C), wet/dirty environment, vertical shaft.

■ Lubrication Programme Essentials

Use colour-coded grease guns and fittings to prevent cross-contamination.

Keep lubrication records in CMMS — track quantity, product, date, technician.

Oil analysis (ferrography, viscosity, water content) at each change confirms condition.

07 Mechanical Systems Maintenance

Rotating mechanical components — bearings, gears, shafts, couplings, belts, and seals — are the most common source of unplanned industrial failures. Systematic inspection and condition monitoring of these components is central to any world-class mechanical maintenance programme.

Bearing Maintenance

- **Installation:** Always use proper fitting tools. Never strike inner ring directly. Heat-fit large bearings to 80–100°C (oil bath or induction heater).
- **Clearance:** Verify internal radial clearance (IRC) matches application. Pre-loaded bearings need precise shim selection.
- **Contamination:** Maintain seal integrity. Contamination causes 14% of premature failures (SKF data).
- **Overheating:** > 70°C ambient on housing indicates over-lubrication, overload, or misalignment.
- **Replacement:** Replace in matched sets on multi-bearing shafts. Record make, type, lot number in CMMS.

Shaft Alignment Standards

Coupling Speed (RPM)	Offset Tolerance (mm)	Angular Tolerance (mm/100mm)	Method
< 1,000	0.10	0.10	Dial indicator
1,000–3,000	0.05	0.05	Laser alignment
3,000–6,000	0.025	0.025	Laser alignment
> 6,000	0.015	0.015	Laser + proximity probe

■ V-Belt Maintenance

Inspect belt tension monthly: deflection = (span/64) in mm under 2N force.

Replace belts in full matched sets — never mix old and new on multi-belt drives.

Misalignment > 0.5° causes rapid belt edge wear and bearing side-loading.

08 Hydraulic & Pneumatic Systems

Hydraulic and pneumatic systems power actuators, presses, lifts, clamps, and control valves across virtually every industrial plant. Contamination control and seal integrity are the two highest-priority maintenance activities for fluid power systems — responsible for over 70% of hydraulic failures.

Hydraulic System Inspection Checklist

- **Fluid level:** Check reservoir level daily. Low level causes pump cavitation.
- **Fluid condition:** Sample and test oil every 3 months — check viscosity, water content, particulate count (ISO 4406 cleanliness).
- **Filter condition:** Replace hydraulic filter at indicator trip or 500-hour interval.
- **Leak inspection:** Zero tolerance for external leaks — repair all seals, fittings, and hose connections.
- **System pressure:** Verify relief valve set-point monthly using calibrated gauge.
- **Actuator speed:** Measure cylinder stroke time — deviation indicates internal bypass or pump wear.
- **Temperature:** Normal operating temperature 40–60°C. > 70°C indicates heat exchanger or valve fault.

Pneumatic System Maintenance

Component	Inspection Task	Interval
Air compressor	Belt, oil, filter, valve check	Monthly
Air dryer/separator	Drain, desiccant, dew-point check	Monthly
FRL units	Filter drain, regulator pressure, lubricator oil	Weekly
Cylinders	Seal integrity, cushion adjustment	Quarterly
Solenoid valves	Response time, seal condition	Annually
Piping system	Leak survey (ultrasonic detector)	Quarterly

■ Contamination Control

Target hydraulic cleanliness: ISO 4406 class 16/14/11 for servo systems.

Use only approved fluid — never mix petroleum and synthetic hydraulic oils.

All new oil must be pre-filtered to target cleanliness before filling system.

09 Electrical Systems Maintenance

Electrical maintenance encompasses the upkeep of power distribution, motor control centres (MCCs), transformers, switchgear, cabling, and instrumentation. Electrical failures are the leading cause of industrial fires and equipment damage — making planned electrical PM and condition monitoring critical safety requirements.

Key Electrical Maintenance Tasks

System	Task	Interval	Technique
LV Switchgear/MCC	Thermography, torque check, clean	Annually	Infrared camera
HV Switchgear	Contact resistance, insulation test	Annually	Ductor, Megger
Transformers	Oil sampling, DGA, thermal scan	6-monthly	Oil lab, IR
Electric motors	Insulation test, air gap, vibration	Annually	Megger, vibration
Cables & terminations	Thermal scan, visual, tug test	Annually	IR thermography
UPS/Batteries	Load test, terminal resistance, capacity	6-monthly	Battery analyser
Earth/Grounding	Earth continuity & resistance test	Annually	Earth tester

Motor Insulation Resistance — Acceptance Criteria

- **Megger Test Voltage:** 500V DC for motors up to 1kV; 1000V DC for 1–6kV motors.
- **Minimum IR (at 40°C):** 1 MΩ per kV of rated voltage + 1 MΩ (IEEE 43 rule).
- **Polarisation Index (PI):** R_{60s} / R_{1min} — PI > 2.0 is acceptable; < 1.0 is critical.
- **Dielectric Absorption Ratio (DAR):** $R_{60s} / R_{30s} > 1.25$ acceptable.
- **Action levels:** IR < 2MΩ → investigate; PI < 1.5 → dry-out or rewind.

■ Electrical Safety Reminder

All electrical work requires Permit to Work (PTW) and verified LOTO isolation.

Never measure insulation resistance on energised conductors.

Document all test results in CMMS against asset history for trend analysis.

10 Maintenance Safety — LOTO & PTW

Lockout/Tagout (LOTO) and Permit to Work (PTW) are the primary safety control systems for maintenance activities. LOTO prevents the unexpected energisation of equipment during maintenance work. PTW is the formal authorisation system that ensures all hazards are identified and controlled before work begins.

LOTO — 6 Step Procedure (OSHA 1910.147)

- **Step 1 — Notify:** Inform all affected personnel that equipment will be isolated.
- **Step 2 — Identify:** Locate all energy sources (electrical, hydraulic, pneumatic, gravity, thermal, chemical).
- **Step 3 — Isolate:** Operate all energy-isolating devices to de-energise the equipment.
- **Step 4 — Lockout/Tagout:** Each worker applies their personal lock and tag to all isolation points.
- **Step 5 — Release stored energy:** Bleed pressure, block gravity loads, discharge capacitors, drain fluids.
- **Step 6 — Verify:** Test for zero energy using calibrated meters before commencing work.

PTW Categories

Permit Type	Required For	Key Control
Cold Work Permit	Non-hazardous mechanical tasks	LOTO verification
Hot Work Permit	Welding, grinding, cutting — ignition risk	Gas test + fire watch
Confined Space Entry	Work inside vessels, tanks, manholes	Continuous air monitoring
Electrical HV Permit	Work on HV systems (> 1kV)	Authorised persons only
Working at Height	Work > 2m above ground level	Full body harness, anchor
Excavation Permit	Digging near buried services	Service drawings + spotter

■ Zero Harm Principles

LOTO applies to ALL energy sources — not just electrical (hydraulics, steam, gravity).

Never share locks. Each technician working on equipment applies their OWN lock.

PTW must be signed off by authorised person before work starts and after completion.

11 Spare Parts & Inventory Management

Spare parts management ensures that the right part is available at the right time, without excessive capital tied up in slow-moving stock. Poor spares management is a top driver of extended Mean Time To Repair (MTTR) — with some studies indicating that 30–40% of downtime is caused by waiting for parts.

Criticality Classification Matrix

Criticality	Criteria	Stock Policy	Lead Time Target
Critical (A)	Single-point failure; long lead time; high cost of failure	Always stock (min/max)	< 24 hours
Important (B)	Significant downtime if unavailable; medium lead time	Stock based on usage	< 1 week
Standard (C)	Short lead time; readily available from local suppliers	Order on demand / JIT	< 2 days
Consumable (D)	Cheap, high-turnover items (filters, seals, gaskets)	Bulk stock with min	Stock carried

Min/Max Inventory Formula

- **Reorder Point (ROP):** Average daily usage × Lead time days + Safety stock
- **Safety Stock:** (Max daily usage – Average daily usage) × Max lead time days
- **Economic Order Quantity (EOQ):** $\sqrt{2 \times \text{Annual demand} \times \text{Order cost} / \text{Holding cost per unit}}$
- **Inventory Turnover:** Annual issues value / Average stock value. Target: 2–4× per year for MRO.

■ Spare Parts Best Practices

Tag every spare with asset number, OEM part number, and shelf life where applicable.

Conduct annual stock-take to identify obsolete, damaged, or expired items.

Use OEM recommended spares lists as the starting point for critical spares strategy.

12 KPIs & Maintenance Metrics

Maintenance Key Performance Indicators (KPIs) provide objective, data-driven insight into the effectiveness and efficiency of the maintenance function. Selecting the right KPIs and trending them consistently allows maintenance managers to identify opportunities, justify investment, and demonstrate value to the business.

KPI	Formula	World-Class Target
Overall Equipment Effectiveness (OEE)	Availability × Performance × Quality	≥ 85%
Asset Availability	$(\text{Total Time} - \text{Downtime}) / \text{Total Time} \times 100$	≥ 95%
Mean Time Between Failures (MTBF)	Total uptime hours / Number of failures	Maximise (trend)
Mean Time To Repair (MTTR)	Total repair time / Number of failures	< 4 hours (critical equip.)
Planned Maintenance Compliance	PMs completed on time / PMs scheduled × 100	≥ 95%
Reactive Maintenance Ratio	Unplanned WOs / Total WOs × 100	< 20%
Maintenance Cost / RAV	Annual maint. spend / Replacement Asset Value	2–5%
Schedule Compliance	WOs completed per plan / WOs scheduled × 100	≥ 90%

OEE Deep Dive

- **Availability Loss:** Breakdowns, changeovers, startup delays → target > 90%.
- **Performance Loss:** Slow cycles, small stops, reduced speed → target > 95%.
- **Quality Loss:** Defects, rework, scrap at startup → target > 99%.

■ KPI Reporting Cadence

Daily: Asset availability, breakdown count, emergency WOs.

Weekly: PM compliance, backlog status, schedule compliance.

Monthly: MTBF, MTTR, OEE, maintenance cost trend, critical spare stock level.

13 Troubleshooting Guide & Standards

Effective fault diagnosis is a systematic process of identifying, isolating, and rectifying equipment failures. The 5-Why and fault tree analysis methods, combined with structured troubleshooting checklists, reduce MTTR and prevent repeat failures by addressing root causes rather than symptoms.

Common Failure Symptoms & Probable Causes

Symptom	Probable Causes	First Action
High bearing temperature	Over-grease, misalignment, overload, bearing defect	Check lube, measure vibration
Excessive vibration	Imbalance, misalignment, bearing wear, looseness	Vibration analysis, laser align
Pump low flow / pressure	Wear rings, impeller erosion, cavitation, bypass	Performance curve test
Motor trips on overload	Overloaded, phase loss, high ambient, winding fault	Check current, insulation test
Hydraulic system slow	Low oil, worn pump, valve sticking, bypass	Pressure & flow test
Air compressor pressure drop	Leaks, worn valves, ring/cylinder wear	Leak test, valve inspection
Gearbox noise (howling)	Gear pitting, incorrect backlash, lubrication failure	Oil sample, visual inspection

Key Standards Reference

- **ISO 55001:** Asset Management Systems — framework for optimising asset lifecycle.
- **ISO 13306:** Maintenance terminology — standard definitions for all maintenance terms.
- **ISO 10816 / 20816:** Mechanical vibration — evaluation of machine vibration severity.
- **IEC 60364:** Electrical installation maintenance requirements.
- **OSHA 29 CFR 1910.147:** Lockout/Tagout (LOTO) standard for hazardous energy control.
- **SAE JA1011/JA1012:** Evaluation criteria and implementation guide for RCM.

■ Root Cause Analysis (RCA)

Use 5-Why analysis for every critical failure — document findings in CMMS.

RCA output: corrective action + preventive action + PM/PdM task update.

Share lessons learned across sites to prevent repeat failures plant-wide.